Mechanisms for Transportation Infrastructure Investment in Developing Countries

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Abstract

In this paper, the authors examine different investment mechanisms for transportation infrastructure projects involving the private enterprise in developing countries. Roles identified vary from those of a financier to an operator for successful public-private ventures. A case study involving such a joint venture in India, the Mumbai Pune Expressway/National Highway 4 (MPEW/NH4) is presented, and fiscal implications of the program, both from the perspective of the public and the private enterprise are examined. The study concludes that if properly planned, joint ventures can be mutually beneficial. A joint public-private program may enable the public sector to use the resources saved for other public projects. It also provides the private agency an opportunity to invest monies in a profitable enterprise that yields social benefits, (e.g. improving mobility, promoting economic development, etc.). Careful analysis must be conducted before the project is undertaken to assess the financial and economic implications of the project from each

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participant's viewpoint, with due regard to risks and uncertainties associated with such long term investments.

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Introduction

The multi billion dollar highway infrastructure in the United States (US) was built over the last 200 years and has been financed primarily by public dollars through various forms of user taxes (Garber and Hoel 2001). Factors such as improved mobility, reduced congestion, and higher safety, along with economic benefits have been used to justify these investments. Tollways and turnpikes, regardless of tenure, constitute a very small fraction of US highways, and are somewhat of an exception to this rule. Typically, these facilities are financed by long-term bonds, and the revenue generated by the facilities is used to pay for the investment. Very little private funding has been used in the past in the US for roadway infrastructure. Private participation is, however, more common in other modes of transportation, particularly rail, air, and transit prior to 1950s. The concept of Public Private Partnership (PPP) in large infrastructure projects that allows joint ownership of infrastructure has gained some ground during the last decade in the US (Rosenau 2000).

The object of interest of this paper is investment in transportation in developing countries in Asia, where infrastructure is as much of a priority as in developed countries to meet their mobility needs and to promote economic development. The discussion in the above paragraph relative to US highways is presented only as a reference. Because of a lack of sufficient capital funding, the private sector is being increasingly called upon to participate in such infrastructure programs in developing countries. India for example, has historically used tax revenues as the primary source of funds for road projects. A recent World Bank Report shows that India currently has 3.5 million kilometers (km) of roads, of which approximately 170,000 km are under the national and state highway category (mostly two-lane facilities), representing modest design standards. The National Highway System in India totaling 58,000 km of two-lane facilities, carries 45% of total traffic (World Bank 2004). In spite of significant public investment

on roads by the Government of India, there is a great need today for high quality, high capacity highways to accommodate the ever-increasing traffic in India's metropolitan areas.

Among major transportation programs that the Government of India has undertaken are: a $50 billion highway improvement for the national and state highway system and another multi billion dollar program to connect small villages through a network of roadways. Additionally, $1.5 billion are needed annually to maintain the 170,000 km of national and state highways that serve as the backbone for regional mobility and economic development. But, resources needed to support the construction and maintenance of this massive infrastructure development are not readily available at the Governmental exchequer facing deficits amounting to 9.5 percent of the country's Gross Domestic Product (GDP)[1] (World Bank 2004).

India has, in the past, used the traditional approach of road financing, where roads are treated as publicly owned/operated facilities, and are funded from a myriad of sources including general revenue, road user taxes, etc. Private sector financing is being sought increasingly to fund infrastructure programs and tolls are being applied to generate revenues. However, private sector financing "cannot replace the role of the public sector, nor can it reduce the importance of rational, fair and transparent public financing system" (World Bank 2004, p. ii). Thus, joint public-private ventures appear to be the key to the financial success of such projects.

The purpose of this paper is two-fold. First, a brief discussion is presented on the emerging roles of the private sector in transportation infrastructure development in India, with specific examples when possible. Second, a case-study on the recently completed Mumbai-Pune Expressway (MPEW), and a companion facility along the National Highway 4 (NH4) is presented, where the private sector has been called upon to play a major role in the long term operation of these two facilities. The financial implications of this role, both to the public and

[1] total market value of all goods and services produced in the country in a given year

private entity are examined and general conclusions drawn about the future roles of the private sector in developing countries in Asia.

Role of the private sector

PPP is a technique to attract private capital in a public project that would otherwise be beyond the reach of the public entity. It has been used both in developing and developed countries as a means of delivering, operating and maintaining infrastructure projects, such as bridges, highways, urban rail systems, and power plants. These projects can be built for new construction to sustain future growth, and for upgrading existing facilities for alleviating congestion, and for rehabilitation efforts. Examples include private sector participation in: the SR-125 project in San Diego County California (Garin et al. 1995), the city of Cleveland for the long term sustainable development (Gauss 2002), a road rehabilitation and expansion project in Orange County California (Henk 1998), a large city link toll road project in Melbourne, Australia (Conde et al. 2008), a light-rail transit system in Portland, Oregon (Landers 2002), tunnel projects in Hong Kong (Zhang and Kumarswamy 2001), a rail transit project in Delhi, India (DMRC 2007) and a series of toll bridge projects in India (Malini 1999).

The aforementioned projects are examples of successful PPP projects that have three common ingredients: identification of the major project participants, a commitment by the participants to address the key issues, and development of strategies to share future risks (Ashley et al. 1998). The first step of a PPP project is to identify key participants such as the eventual owner, developer, financial investor, and the community users. The second step is to develop a common forum of understanding of the diversity of interests among the project participants, thereby formalizing policy issues. The third step is to formulate a robust approach to share risks that may be encountered during the life of the project.

The literature review conducted as a part of this research indicates a number of ways in which the private sector can be involved in transportation infrastructure investment. Discussions with experts in India indicate that the private sector has been involved in two ways in India: as a lender and as a promoter.

The Private Sector as a Lender

In this role, the private entity acts as a financier and is interested in the project primarily from an investment point of view, where profit margins are expected to be high. Depending on the size of the project, the private entity can be a single investor; or a group of investors; a corporation; a banking institution or a group of corporations/institutions. The private entity, depending on the agreement, may or may not participate in the policy-making activities of the Principal/Organization seeking the loan.

An example of this role in India can be found in the urban rail system currently being built in Delhi Metropolitan Area (the nation's capital) by the Delhi Metropolitan Rail Corporation (DMRC). This is a multi-phase urban rail transit project designed to meet projected traffic demand for the year 2021, consisting of 67 stations and 244 km of rail network to be built in four phases. Phase 1 of the project has recently been completed and includes a combination of underground, elevated and at-grade sections comprising 65 km and 59 stations (DMRC 2007). A group of Japanese Banks, led by Japan Bank for International Corporation (JBIC), is a major financier for the project, and is expected to cover more than 60 percent of the project cost (Sreedharan 2004).

The Private Sector as a Promoter

A promoter may participate under the Build Own Operate Transfer (BOOT) program (or variations thereof), raise capital funds from private resources and build and operate the facility in

exchange of future revenues to be generated by the project. It has been used extensively in Europe and more recently in India and is defined as (Meerna and Njiru 1998, p. 79):

> "A project based on the granting of a concession by a Principal, usually a government, to the Promoter, sometimes known as the Concessionaire, who is responsible for the construction, financing, operation and maintenance of a facility over the period of concession before finally transferring the facility, a no cost to the Principal, as a fully operational facility. During the concession period the Promoter owns and operates the facility and collects revenues to repay the financing and investment costs, maintains and operates the facility and makes a margin of profit."

The concession agreement defines the roles and responsibilities of the participating agencies, particularly the Principal, the Promoter, and the support agencies. BOOT projects are essentially turnkey contracts financed by the contractor, with extended operation and maintenance periods. If the project is planned properly, the principal or the Government agency has nothing to lose, as it essentially inherits a "fully operational facility," free and clear, at the end of the concession period. It is, however, important for the Governmental agency to ensure that the facility continues to generate revenue at the end of the concession period without a major infusion of resources. The private entity on the other hand, can take advantage of an investment opportunity, and generate a healthy return over the concession period.

Variations of the BOOT concept have been used in different countries in many different forms:

- FBOOT finance-build-own-operate-transfer;
- BOO build-own-operate;
- BOT build-operate-transfer;
- BOL build-operate-lease;

- DBOM design-build-operate-maintain;
- DBOT design-build-operate-transfer;
- BOD build-operate-deliver;
- BOOST build-own-operate-subsidies-transfer;
- BRT build-rent-transfer;
- BTO build-transfer-operate;
- ROT rebuild-operate-transfer
- LOT Lease-operate-transfer

Among the concepts outlined above, the last two, ROT and LOT have been used in India in a number of recent projects. The rebuilding part of the ROT program can be in the form of upgrading the facility by improving the pavement, geometric/operational features; and/or expanding the current facility by such actions as: adding lanes, adding shoulders, building grade-separated structures, etc. The private entity is allowed to operate the facility and to collect toll charges during the concession period, and thereby recover its capital investment. The public entity is spared the investment cost as well as the task of operating the facility during the concession period. The case of NH4 connecting Mumbai with Pune discussed later in case study, is an example of this program.

The LOT program is another derivative of the BOOT concept, where the private entity leases an existing facility for a specified amount; operates the system for the concession period; and transfers the operational rights of the facility to the Principal at the end of the prescribed period. Technically, the public entity “sells” the operational rights of the facility to the private agency for a negotiated price. In exchange, the private agency “earns” the right to collect and retain the toll charges as its earnings from the investment. The facility must be returned to the government in fully operational condition at the end of the prescribed term. The MPEW project discussed in the next section is an example of the LOT concept.

Methodology

The methodology used in this study is presented in two sections: (1) deriving the Internal Rate of Return (IRR), and (2) procedure used for estimating costs and benefits.

IRR

IRR provides an estimate of the return or yield of the investment, given a set of expenditure and revenue data along with their expected dates over the life of the project. It is the annualized compounded return rate derived from an investment comprising payments and earnings at different points/periods in time during the tenure of the project. IRR is defined as the interest rate at which the Net Present Worth (or Net Annual Worth or Net Future Worth) of the investment is equal to zero. The generic form of the model is:

$$C_c + P_{om} + R_{om} = B_t + B_{voc} + B_{tt} + B_{sa} \qquad (1)$$

where,

C_c : present worth (PW) of construction cost incurred prior to the implementation of the project

P_{om}: PW of periodic operation and maintenance (POM) incurred in different points in time

R_{om}: PW of regular operation and maintenance (ROM) incurred in every year

B_t : PW of benefit accrued from toll receipts in different years

B_{voc}: PW of benefit accrued from vehicle operating cost savings in different years

B_{tt} : PW of benefit accrued from travel time savings in different years

B_{sa} : PW of the amount received by "leasing the facility" to operate during the concession period

Since the cash flow elements described above are incurred at different points / periods in time, appropriate interest factors are to be applied to convert them to their PW equivalent. The specific interest factors would depend upon the exact nature and time frame of the cost/benefit items and are not shown in the generic model in Equation 1. The procedure to estimate '*i*' is

heuristic one, by which the '*i*'-value is sought at which the NPW approaches a zero value. This is accomplished by systematically changing the value of '*i*' in Equation 1 until convergence is reached. The algorithm used symbolically is:

Estimate the interest rate '*i*' at which,

The Net Present worth (NPW) of the investment = 0, or,

Present worth of Costs = Present worth of Benefits

Further, the IRR derived using Equation 1 must be at least equal to the Minimum Attractive Rate of Return (MARR), where MARR is the rate of return below which the investment proposal is to be deemed unacceptable (Blank and Tarquin 2005). The MARR value is based upon predominant lending rates and economic conditions, and is often treated as an exogenous variable, a detailed discussion of which is beyond the scope of this paper. It is however conceivable, that the public and private entities may identify different MARR's as a benchmark of accepting/rejecting investment proposals.

Economic Internal Rate of Return (EIRR) vs. Financial Internal Rate of Return (FIRR)

The IRR technique has been used in the literature to evaluate project viability. Examples include decision economic model for parking facility planning in urban locations (Merino 1989), ranking of the transportation projects by their financial rate of return (Bonnafous and Jensen 2005), use of IRR for a series of economic evaluation projects (Borgonovo and Peccati 2005), and testing the feasibility of multiple agency development projects at proposed transit stations (Khasnabis et al. 1982).

For joint public private projects, questions are often raised about the inclusion of externalities, or social costs/benefits. Externalities include environment damages, pollution, savings in travel time, travel cost, etc., that are not reflected in the markets (Johnson and Kasarda 2007). It is customary to ignore these externalities in any analysis dealing with the private sector,

as these do not affect its decision. The term used for this return is Financial Internal Rate of Return (FIRR), where only direct expenditures and revenues are included. For the public sector, two sets of returns are generally estimated. First, the FIRR is used to benchmark public sector performance with that of the private sector. An additional analysis that includes the externalities (both costs and benefits) is conducted to estimate the Economic Internal Rate of Return (EIRR).

The definitions of EIRR and FIRR are as follows:

EIRR: The rate of return that would be achieved on all project resource costs, where all benefits and costs are measured in economic prices. For a project to be acceptable, the EIRR should be greater than the economic opportunity cost of capital (ADB 2000).

FIRR: The rate of return that would be achieved on all project costs, where all costs are measured in financial prices and when benefits represent the financial revenues that would accrue to the main project participant. It should be compared with the Opportunity cost of capital, or the weighted average cost of capital, to assess the financial sustainability of a project (ADB 2000).

In this paper, two broad categories of analysis are reported, (1) for the private entity, and (2) for the public entity. Following the principles described above, for the private entity, the only benefit considered were the toll receipts for FIRR analysis. For the public entity, benefits include only the toll receipts for FIRR analysis. Additionally, savings in vehicle operating cost (VOC) and savings in travel time (TT) were included in the EIRR analysis. These savings were compiled based upon a "Do-Nothing Alternative." Referring to Equation 1, in computing FIRR, the values of B_{voc} and B_{tt} were assumed to be zero. For computing EIRR, numerical values of B_{voc} and B_{tt} were derived as explained in the next section and included in Equation 1.

Procedures for Estimating Costs and Benefits

While the data on toll charges, volume, expected growth, etc., were derived directly from MSRDC, savings in VOC were computed utilizing a comprehensive procedure developed by the Central Road Research Institute (CRRI), one of the 19 research institutes supported by the Council of Scientific and Industrial Research (CSIR), Government of India (CRRI 2001). A detailed discussion of this procedure is beyond the scope of the paper. This procedure consists of a set of sequential steps of deriving parameters through the use of models calibrated with Indian roads and traffic conditions. The procedures used to estimate VOC and TT are summarized below.

VOC

The VOC values for each mode are determined by the pavement roughness, rise and fall, and the carriageway width available for movement of traffic for various road types. VOC per kilometer can be estimated as:

$$log_e^{VOC^{m_i}} = c_1 + c_2 R_t + c_3 RF + c_4 W \qquad (2)$$

Where

R_t: roughness at time period t

RF: rise and fall in meters / km

W: width of carriageway in meters

$c_{1...}$ c_4: calibrated parameters to estimate vehicle operating cost

Further, roughness contains other deterioration characteristics such as cracking, rutting, patching, raveling and potholing. The roughness model adapted from CRRI is described below (CRRI 2001).

$$R_t = R_0 + \phi N_t \qquad (3)$$

$$\phi = \frac{1250}{anti\,log_{10}\left(a^{1/3} - b^{1/3} + 1.3841\right)} \tag{4}$$

$$a = 0.20209 + 23.1318C^2 - 4.809C \tag{5}$$

$$b = 0.20209 + 23.1318C^2 + 4.809C \tag{6}$$

$$C = 2.1989 - MSN \tag{7}$$

$$MSN = \sum_{\omega=1}^{N} a_\omega D_\omega + 3.51 \times log_{10}^{CBR} - 0.85\left(log_{10}^{CBR}\right)^2 - 1.43 \tag{8}$$

$$N_t = 10^{-6} \sum_{m=1}^{M} y_t^m \gamma_f^m l_d \tag{9}$$

Where,

R_0: initial roughness

N_t: number of millions of standard axles at time period t

a,b, and C: parameters to calculate ϕ

MSN: modified structural number

ϕ: estimated coefficient for calculation of R_t

a_ω: the strength coefficient for layer ω

D_ω: the thickness of layer ω in inches

N: total number of layers in the pavement

CBR: California Bearing Ratio of the subgrade

y_t^m : annual traffic volume for mode m at year t

γ_f^m : truck factor for mode m

l_d : lane distribution factor

Equation 3 is used to obtain the roughness of the pavement surface. The values of initial roughness, number of millions of standard axles and its coefficients are required to calculate the parameter ϕ represented in Equation 4. Equations 5-7 are applied to estimate ϕ. Equation 7 requires Modified Structural Number (MSN) to calculate *C*. The MSN (Equation 8) is a pavement characteristic which is a function of depth of pavement layers, layer coefficients and

California Bearing Ratio (CBR). Equation 9 determines the number of Millions of Standard Axles (MSA), to be used in Equation 3. Traffic consists of various modes; so truck factors are used to convert all modes to their equivalent number of standard axles. The annual traffic volume is multiplied by corresponding truck factors (for Indian conditions obtained from CRRI) and lane distribution factor to obtain MSA.

TT

Monetary value of travel time incurred for each mode to traverse a facility can be estimated as:

$$T_t = \sum_{m=1}^{M} v_t^m o_c^m t^m \qquad (10)$$

Where:

T_t : total value of travel time for all modes at the end of year *t*

v_t^m : value of travel time per hour for mode *m*

o_c^m : occupancy of mode *m*

t^m : travel time needed to traverse the facility for mode *m*

TT saved for a proposed facility can be calculated as the difference between the T_t – values between the existing facility and the proposed facility.

Case study of MPEW and NH4

Mumbai is the commercial and financial capital of India with a population of more than 15 million. Pune with a population of over five million is a major urban center in the state of Maharashtra, and is growing into a major industrial and commercial center, being the automotive capital of India. Hence, the importance of Mumbai-Pune travel corridor increased tremendously in last decade. The travel demand is currently served by a multimodal system comprising rail, air

and highways. The road traffic demand warrants a ten-lane system between the two cities (MSRDC 2007).

Background Information

For a long time, the National Highway 4 (NH4) was the only available roadway connecting the cities of Mumbai and Pune. This section of NH4, a two-lane roadway built to modest design standards, is a part of the National Highway System and is one of the most congested facilities in the country. The poor operating condition on NH4 made it necessary for the Government of Maharashtra to build an independent toll expressway, as this travel corridor represents the economic backbone of the region. The Maharashtra State Road Development Corporation (MSRDC), an agency of the state Government built the MPEW, a six-lane facility with high design standards in 2000 and was authorized to collect tolls from road users (MSRDC 2007). MSRDC has invested 2100 crores[2] of Indian rupees (the equivalent of $525 million at the rate of 40 Indian rupees per US dollar), over a four year period to construct the 95 km long six-lane carriageway using high design standards and modern machineries. MPEW was opened to traffic as a toll facility in the year 2001 (Fig. 1). A foreign exchange conversion rate[3] of 40 Indian rupees per US dollar was used in this paper (as of March 31, 2008, when the analysis was conducted).

In the year 2004, MSRDC entered into an agreement with a private entity to operate, maintain and collect tolls on MPEW and to rebuild, maintain, operate and collect tolls on NH4, both for a period of 15 years (2005-2019). Further, the private operator paid MSRDC $225 million (900 crores of Indian rupees) and agreed to invest necessary capital to upgrade NH4 to a four-lane facility. The MPEW and the NH4 projects can thus be designated as a LOT and a ROT project respectively, in the light of the discussion presented earlier.

[2] Note: 1 crore = 10 million
[3] Source for foreign exchange rate: http://www.x-rates.com/cgi-bin/hlookup.cgi

During the lease period, the private entity was required to widen NH4 from two lanes to four lanes and to open the facility to traffic in 2007 (Fig. 1). MPEW and NH4 are to be delivered back to MSRDC and the Government of Maharashtra respectively in 2020 in fully operational condition. MSRDC is expected to resume its role of operating and maintaining MPEW for the remainder of its service life up to the year 2030. Since NH4 is an older facility, it was assumed for the purpose of this case study, that NH4 will not have any further life left after 2019. Realistically however, to meet the traffic demand, the Government may be required to make a significant investment to keep the facility operational. No decision on the disposition of NH4 beyond 2019 has been made at this point.

Table 1 shows the traffic and other key features of the two facilities obtained mostly from MSRDC. Six categories of vehicles with different toll rates and operating characteristics were considered: Car, Light Commercial Vehicles (LCV), Small Truck, Bus, 3 Axle (3AX) and Multi Axle (MA) trucks. LCV's, small trucks, 3AX and MA trucks are generally used for freight transportation. As explained earlier, the equivalent $525 million (2100 crores of Indian rupees) was invested in the MPEW facility during the period 1997-2000. Per MSRDC estimates, approximately 20 percent of the capital cost of $525 million was incurred, each at the end of the first and fourth year and 30 percent was incurred, each at the end of the second and third year. The cost of improving NH4 to a four lane facility was estimated as the equivalent of $100 million, mostly invested in 2004.

Results

The base year savings in VOC and TT (for the two candidate alternatives, MPEW and NH4 with respect to the NH4 (old) or the base condition)), computed according to the CRRI procedures are presented in Tables 2 and 3 respectively. In Table 2, various parameters are

computed for the three facilities (MPEW, NH4 and NH4 (old)) for five vehicular categories, and presented in columns 3, 4, 5, and 6. The mode-specific VOC's for each facility computed using Equations 2 through 9 are presented in column 7. The values of RF, MSN, CBR, and other parameters were obtained from CRRI, and numerical values of other variables were obtained from MSRDC. The savings in VOC for MPEW and NH4 are obtained as the difference between the base case and the facility, and presented in column 8 of Table 2. For example, savings in VOC for cars traveling on MPEW are calculated as $4.010 per 100 km, being the difference in the VOC for base condition (7.043) and for MPEW (3.033). Annual savings in VOC for each facility are estimated as the sum-product of traffic volume for each mode and the corresponding VOC savings (column 9). The operation cost for each facility is obtained as the sum-product traffic volume for each mode and the corresponding VOC presented in column 7. The operation cost for the year 2001 is shown in column 10 of Table 2. Maintenance cost[4] is estimated (column 11) by multiplying a standardized value of 10,000 Indian rupees ($250) per lane per km per year of maintenance cost, number of lanes (see Table 1), and the length of the facility (see Table 1). Total operation and maintenance cost is presented in column 12 is the summation of operation cost (column 10) and maintenance cost (column 11). In Table 2, costs and savings are shown for the base year. Similar cost figures (annual) for the entire analysis period were derived using the traffic growth data for each mode presented in Table 1.

Travel time savings data in minutes for each mode is computed as the difference in travel time between the base case and the proposed facility (Table 3, column 3). Unit value of travel time is obtained from CRRI for each mode (Table 3, column 4). The value of travel time saved for each mode is presented in column 5 (CRRI 2001). For example, the value of travel time saved for cars traveling on MPEW is $4.06 per trip, i.e. (75*3.25)/(60). The total value of travel

[4] $Mc = ql_n d$, where, M_c= maintenance cost, q = maintenance cost in rupees per lane per kilometer per year, l_n: number of lanes available for vehicular operation, *d*: total distance of roadway in *km*.

time saved (column 6) for a facility is the sum-product of traffic volume for each mode and the corresponding value of travel time saved.

Public Entity Perspective

Fig. 2(a) and 2(b) show the cash flow diagrams reflected in the economic analysis for the MPEW and NH4 projects respectively from the public entity's perspective. Corresponding results are presented in Table 4. Case (a) of Table 4 shows that the FIRR and EIRR generated for the MPEW project for the public entity are 5.07 percent and 13.95 percent (columns 10 and 11) respectively. Corresponding figures for NH4 are shown in case (b). Since the public entity did not invest any money on NH4 rebuilding, and does not receive any toll from NH4, the FIRR calculation is not applicable (NA). By the same token, the public entity derives the savings in VOC and in TT without any investment, so that corresponding EIRR is very high or infinity.

A number of other analyses are presented in Table 4. Case (c) shows the analysis for the hypothetical case of the public entity continuing to operate MPEW without any private investment whatsoever. In that case, the expected FIRR and EIRR for the public entity would have been 7.05 percent and 14.75 percent respectively, both considerably higher than their corresponding values of 5.07 percent and 13.95 percent under current operation. This is expected, as the public entity must be ready to share the project returns with its partners to attract private capital. The differential between the two sets of EIRR's/FIRR's can be looked upon as the "price" paid by the public entity to secure private participation.

Case (d), Table 4 shows that if the current operating plan for MPEW were to be continued over the life of the project, (i.e. if the private entity is allowed to continue the operation of MPEW beyond the year 2019, up to the year 2030), the minimum "asking price" by the public entity will be $524.31 millions in the year 2019 (column 6) for it to earn the same FIRR of 5.07 percent. Stated differently, the private entity must be wiling to pay a minimum of

$524.31 million to the public entity to continue to operate MPEW and collect toll beyond the year 2019 up to the year 2030.

Case (e) of Table 4 shows the combined FIRR and EIRR to the public entity from the two plans under current operating plans are 5.07 percent and 16.32 percent respectively. Since the public entity neither invests any money in the NH4 rebuilding project, nor receives any direct benefit, the combined FIRR to the public entity from the two projects is the same as the FIRR from MPEW. However, the combined EIRR is higher because of the additional external benefits (savings in VOC and TT) at no extra cost. Lastly, case (f) shows that a combination of current NH4 operation and the hypothetical scenario of MPEW, without any private involvement would have resulted in 7.05 percent and 17.89 percent of FIRR and EIRR respectively for the public entity. As expected, for case (f), the FIRR is the same as in case (c), while the EIRR has increased from 14.75 percent to 17.89 percent.

Private Entity Perspective:

Fig. 2(c) and 2(d) represent the cash flow diagrams from the private entity's perspective for the two facilities as currently planned and implemented. The corresponding results are shown in Table 5. Since no external benefits are involved (being the private entity's perspective), only FIRR values are calculated as 12.77 percent and 33.05 percent for MPEW and NH4 respectively, with a combined FIRR of 20.21 percent. The NH4 project generates much higher return to the private entity than the MPEW project, because it requires much smaller investment, and provides much higher comparative benefit.

Sensitivity and Scenario Analysis

An additional set of analyses was conducted to test if the procedure/methods used to estimate the project returns from the perspective of public and private entity are sensitive to the changes in key variables. Table 6 shows that an increase in operating cost, reduction in volume,

and reduction in toll rates, considered one factor at a time, will result in a reduction in the EIRR and FIRR (where appropriate), both for the public and private entity for the two facilities. Also, Table 6 shows that an increase in toll rate will result in an increase in EIRR and FIRR.

Additionally, a set of scenario analysis, with each scenario comprising changes in all three factors together (operation and maintenance cost, traffic volume and toll rates) is reported in Table 7. Table 7 shows that resultant changes in the EIRR and FIRR do follow expected trends. For example, the combined effect of an increase in O&M cost, a decrease in traffic volume and a decrease in toll rate (Scenarios 1, 2, and 3) is a decrease in EIRR and FIRR. Similarly, increases in traffic volume and in toll rates (Scenario-4) result in an increase in FIRR and EIRR., suggesting that the positive effect of toll increase outweighs the adverse effect of higher volume (resulting in higher operation and maintenance cost).

Conclusions

Initial discussions presented in this paper identify various roles that the private sector may play in supporting transportation infrastructure programs in developing countries. This participation can take a number of different forms, and examples of such private involvement are cited in the paper. A case study involving the MPEW and NH4 in India is presented to examine the consequence of joint private-public participation in a major infrastructure project. The authors feel that the trends, rather than the numbers, are more important in drawing meaningful conclusions regarding the outcome of such projects. The sensitivity and the scenario analyses show that the trends observed are reasonable and logical.

The case study indicates that the PPP infrastructure project has been mutually beneficial. The private entity in this case, derives a healthy return exceeding 20 percent from the combination of the two projects MPEW and NH4, with a larger contribution by the latter project.

The public entity, by transferring the operational rights of the two facilities, loses the opportunity for earning a sustained level of revenue. However even with the income foregone through future toll charges, the public sector earns a 5.07 percent FIRR and 16.32 percent EIRR. Considering that a primary mission of the public sector is to derive social benefits from public investment, a 16.32 percent EIRR appears to be an excellent investment. The $225 million (900 crores of Indian rupees) received from the private entity for the MPEW project, along with the maintenance/operational expenses saved, will allow the public entity to invest public funds in other socially desirable programs. Similarly, the $100 million (400 crores of Indian rupees) "saved" by the public entity along with associated maintenance and operating expenses in rebuilding and operating NH4 can be used to finance other programs to benefit the taxpayers. Lastly, the public entity is protected from any risks/uncertainties associated with collection of future tolls during the concession period.

Additional efforts should be undertaken to examine the financial implications of these programs from the perspective of the public sector and the private sector. Future research should incorporate the concept of risk and uncertainty into the economic analysis. The EIRR and FIRR-values computed in this paper are based upon the assumption of the future returns (toll charges, savings in VOV, and TT) being fully known. In reality, there are uncertainties associated with these returns. Further, over an extended life period, the more distant the future is, the more uncertain the return is likely to be. This feature underscores the importance of incorporating risks and uncertainties into the analysis.

Assuming that the future toll revenues and operating/maintenance expenditures can be predicted with reasonable accuracy, PPP appears to be a very desirable way to support major infrastructure projects in situations where there is not enough capital funding at the public exchequer to support these projects. While this may be true for developing countries in Asia (and

elsewhere), the PPP approach is being increasingly looked upon as a viable tool for supporting infrastructure projects in developed countries as well. Such participation can be mutually beneficial in that it can save the public agency capital expenditures and protect it from future risks associated with raising enough revenues. In exchange for these savings in capital cost and opportunity costs, the public sector is required to forgo future benefits. The private agency has the opportunity to derive a healthy return from the investment, if the facility is maintained properly and operated efficiently. The last factor may serve as an incentive to the private agency to keep the facility under proper maintenance to the benefit of the public agency. The public agency should maintain some degree of control over the operation of the facility during the concession period, and should have a strong voice in the governance of the facility, since the facility is for public benefit after all.

A comprehensive set of financial and economic analyses should be conducted by each participating entity independently before the entity decides to participate in the project in a designated role. Risks and uncertainties should be taken into account to the extent possible, and each participant should make its own decision based upon the information available.

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Acronyms	Explanation
3AX:	three axle vehicles
BOOT:	build own operate and transfer
CRRI:	Central road research institute
DMRC	Delhi Metro Rail Corporation
EIRR:	Economic Internal rate of return
FIRR:	Financial Internal rate of return
IRR:	Internal rate of return
LCV:	Light Commercial Vehicles
LOT:	Lease operate and transfer
MA:	Multi axle vehicles
MARR:	Minimum Attractive Rate of Return
MMRDA:	Mumbai Metropolitan Region Development Authority
MPEW:	Mumbai Pune Expressway
MSN:	Modified Structural Number
MSRDC:	Maharastra State Road Development Corporation
NH4:	National Highway No-4
NMSEZ:	Navi Mumbai Special Economic Zone
RITES	Rail India Technical and Economic Services
ROT:	Rebuild operate and transfer
VOC:	Vehicle Operating Cost

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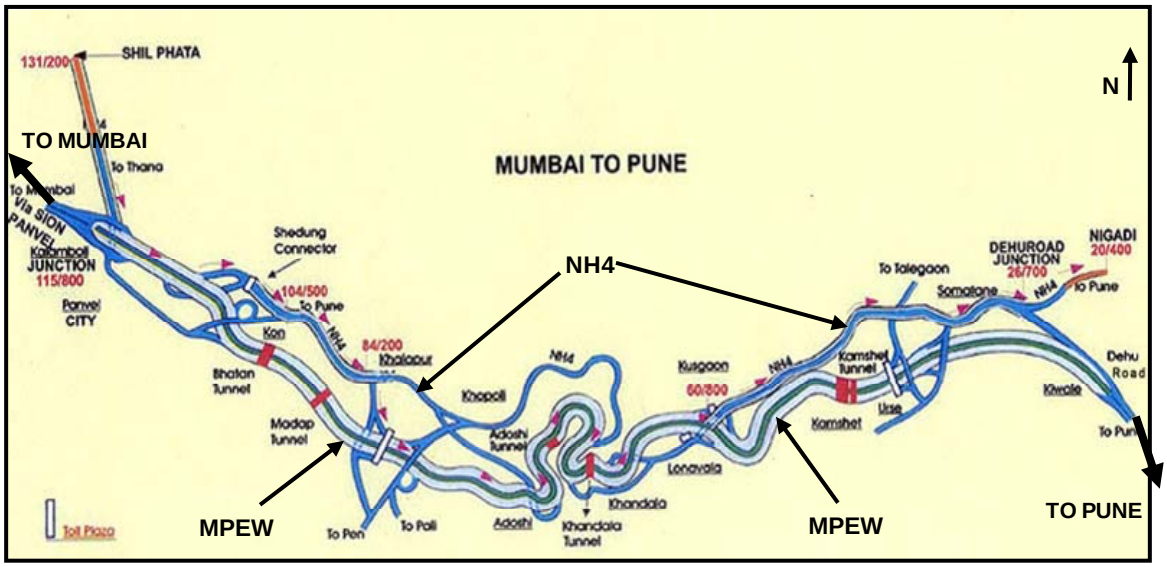
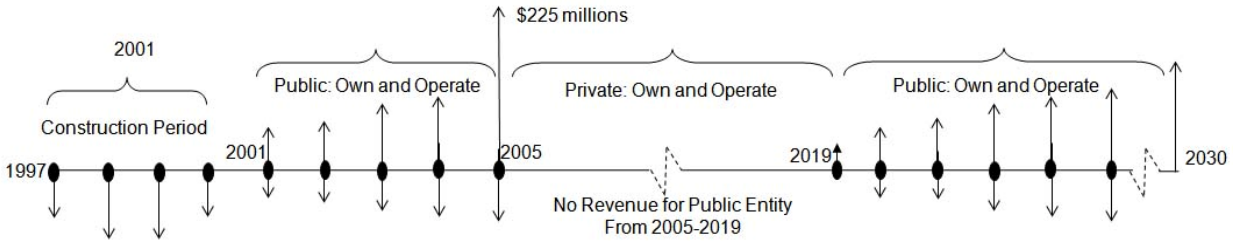
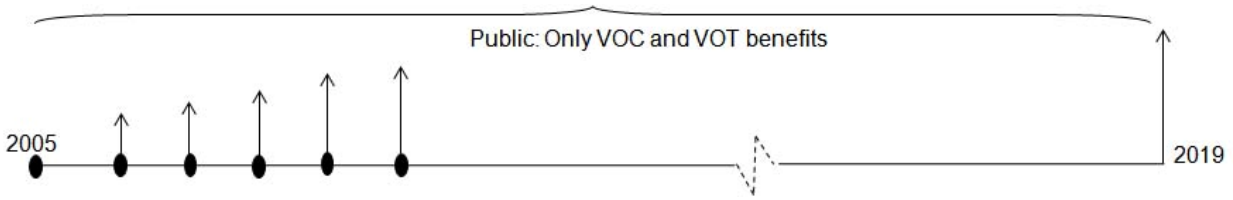


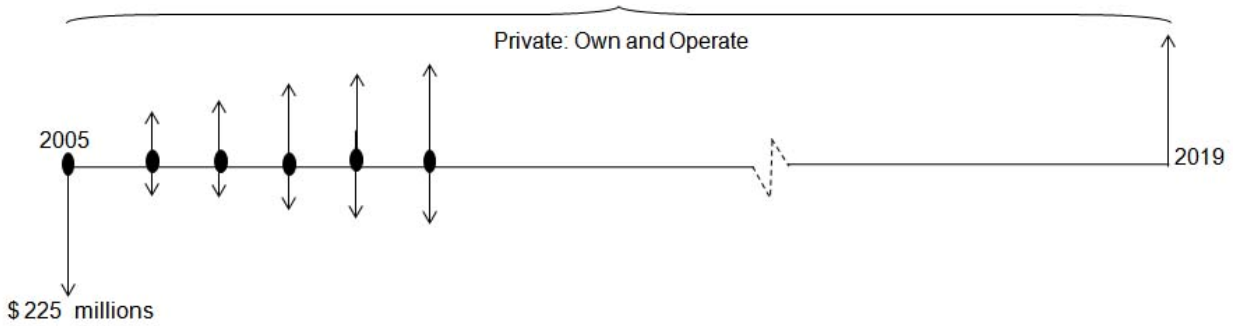
Fig. 1. MPEW and NH4 between Mumbai and Pune (MSRDC 2007)



(a) Economic and financial aspects of the public entity for MPEW-Existing Scenario



(b) Economic and financial aspects of the public entity for NH-4



(c) Financial aspects of the private entity for MPEW

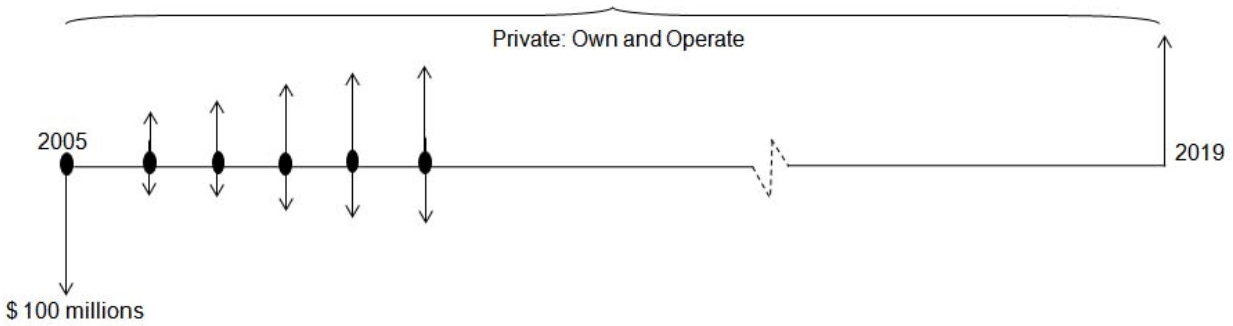


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Table 1. General Information on Alternative Characteristics

Facility	Traffic Volume		Traffic Growth (%)		Speed Limit (km/hr)		Toll Rate ($)	
	MPEW (2001)	NH4 (2005)	MPEW	NH4	MPEW	NH4	MPEW	NH4
Car	3,071,765	2,825,210	5.7	2.73	70	55	3	2
LCV	337,116	1,253,569	9.47	2.65	60	50	5	3
Truck	1,221,929	1,238,703	-8.08	1.94	60	45	6	4
Bus	441,184	628,216	0.07	2.78	60	45	9	6
3AX	117,142	204,481	1.45	7.47	60	45	15	10
MA	44,398	168,984	3.38	20.89	60	45	20	13

Table 2. Savings in Vehicle Operating Cost for the Base Year

FACILITY	MODE	(c_1)[a]	(c_2)[a]	(c_3)[a]	(c_4)[a]	(VOC)[b] $/100km	Savings in VOC ($/100km)	Savings in VOC (Million $)	Operation Cost (Million $)	Maintenance Cost (Million $)	Total O&M Cost (Million $)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
MPEW	Car	0.082	0.038	0.008	NA[c]	3.033	4.010	31.42 (2001)	7.29 (2001)	0.14 (2001)	7.43 (2001)
	LCV	0.618	0.027	0.010		5.069	7.759				
	Truck	0.886	0.024	0.006		6.515	10.037				
	Bus	0.775	0.021	0.009		5.833	8.718				
	3AX & MA	1.380	0.017	0.014		10.651	14.725				
NH4 (New)	Car	0.336	0.056	0.010	NA[c]	4.828	2.215	32.43 (2005)	5.71 (2005)	0.10 (2005)	5.81 (2005)
	LCV	0.748	0.027	0.012		6.822	6.007				
	Truck	1.176	0.031	0.009		10.148	6.403				
	Bus	0.956	0.031	0.010		8.270	6.281				
	3AX & MA	1.384	0.022	0.013		12.940	12.435				
NH4 (Old) (Base Condition)	Car	1.381	0.083	0.013	-0.115	7.043	NA[d]	NA[d]	NA[e]	NA[e]	NA[e]
	LCV	1.643	0.027	0.013	-0.039	12.828					
	Truck	2.072	0.041	0.010	-0.065	16.552					
	Bus	1.935	0.046	0.012	-0.070	14.551					
	3AX & MA	2.726	0.030	0.014	-0.050	25.376					

Note: [a]: c_1, c_2, c_3, and c_4 are obtained from CRRI
[b]: VOC computed from Equation 2
[c]: The coefficient c_4 is zero for four lane (and / or higher) divided highway facility (CRRI 2001)
[d]: Savings are compared to the NH4 (Old) (Base Condition)
[e]: Cost computation for the NH4 (Old) is not applicable

Table 3. Value of Travel Time Savings

Facility	Mode	Travel time savings per trip (min)	Unit value of travel time ($/hr)	Value of travel time saved per trip ($)	Total Value of Travel Time Saved (Million $) for the base year
(1)	(2)	(3)	(4)	(5)	(6)
MPEW	Cars	75	3.25	4.06	19.79 (2001)
	LCV	50	1.50	1.25	
	Buses	60	8.75	8.75	
	HCV	70	1.88	2.19	
	3A	70	1.88	2.19	
	MA	70	1.88	2.19	
NH4 (New)	Cars	50	3.25	2.71	15.33 (2005)
	LCV	33	1.50	0.83	
	Buses	40	8.75	5.83	
	HCV	47	1.88	1.46	
	3A	47	1.88	1.46	
	MA	47	1.88	1.46	

Table 4. Economic and Financial Aspects of the Public Entity

Case	Project	Cost: Initial Cost (Million $)	Cost: Periodic Investment at every 5th year (Million $)	Cost: Regular Operation and Maintenance Cost (1st Year) (Million $)	Project Life	Benefit: Selling Amount (Million $)	Benefit: Savings in VOC (1st Year) (Million $)	Benefit: Savings in VOT (1st Year) (Million $)	Benefit: Toll (1st Year) (Million $)	FIRR (%)	EIRR (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(a)	MPEW (Current operation)	525 (1997-2001)	52.5 (2001)	7.43 (2001)	2001-2030	225 (2005)	31.4 (2001)	19.78 (2001)	24.85 (2001)	5.07	13.95
(b)	NH4 (Current operation)	-	-	-	2005-2019	-	32.42 (2005)	15.32 (2005)	-	NA	Very High
(c)	MPEW (No private involvement)	525 (1997-2001)	52.5 (2001)	7.43 (2001)		-	31.4 (2001)	19.78 (2001)	24.85 (2001)	7.05	14.75
(d)	MPEW (Continued private operation)	525 (1997-2001)	52.5 (2001)	7.43 (2001)		524.31 (2019)	31.4 (2001)	19.78 (2001)	24.85 (2001)	5.07	13.95
(e)	Combination (a + b)									5.07	16.32
(f)	Combination (b + c)									7.05	17.89

Note:

a, b: Represents current operation

c: represents a hypothetical scenario, where the private sector is not involved at all, and the public entity is responsible for all expenditures and system operation

d: represents a hypothetical scenario, where the private sector is responsible for all expenditures and system operation, but it provides one time payment at two stages 2005 and 2019

Table 5. Financial Aspects of the Private Entity

Project	Initial Cost (Million $)	Periodic Investment at every 5th year (Million $)	Regular Operation and Maintenance Cost (Million $)	Project Life	Toll for first year (Million $)	FIRR (%)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
(a) MPEW	225.00 (2005)	52.50 (2005)	7.44 (2005)	2005-2019	30.97 (2005)	12.77
(b) NH4	100.00 (2005)	10.00 (2005)	5.81 (2005)	2005-2019	22.58 (2005)	33.05
(c) Combination						20.21

Table 6. Sensitivity Analysis for both Private and Public Entity

Variation	Percentage Change	Private Entity (3)			Public Entity (4)					
		FIRR (%)			FIRR (%)			EIRR (%)		
(1)	(2)	MPEW	NH4	Cumulative	MPEW	NH4	Cumulative	MPEW	NH4	Cumulative
Base Case	0	12.77	33.05	20.21	5.07	NA	5.07	13.95	Very High	16.32
	10	9.74	28.7	16.93	4.49	NA	4.49	13.37	Very High	15.32
Increase in O&M Cost	20	9.22	28.12	16.87	4.4	NA	4.4	13.32	Very High	15.27
	-5	10.25	29.57	17.52	4.58	NA	4.58	13.41	Very High	15.38
Volume of Traffic	-10	9.25	28.07	16.32	4.35	NA	4.35	12.98	Very High	14.86
	-5	11.36	31.07	18.7	4.8	NA	4.8	13.83	Very High	15.88
	-10	9.91	29.1	17.16	4.52	NA	4.52	13.71	Very High	15.66
Toll Rates	5	14.14	35.04	21.7	5.32	NA	5.32	14.07	Very High	17.16

Table 7. Scenario Analysis for both Private and Public Entity

Scenarios	Variation	Percentage Change	Private Entity (4)			Public Entity (5)					
			FIRR (%)			FIRR (%)			EIRR (%)		
(1)	(2)	(3)	MPEW	NH4	Combination	MPEW	NH4	Combination	MPEW	NH4	Combination
Base Case	No Variation	0	12.77	33.05	20.21	5.07	NA	5.07	13.95	Very High	16.32
	Increase in O&M Cost	10									
Scenario-1	Volume of Traffic	-10	8.3	26.98	15.49	4.21	NA	4.21	12.91	Very High	14.38
	Toll Rates	-5									
	Increase in O&M Cost	10									
Scenario-2	Volume of Traffic	-10	9.66	28.74	16.9	4.48	NA	4.48	13.03	Very High	14.59
	Toll Rates	0									
	Increase in O&M Cost	20									
Scenario-3	Volume of Traffic	-10	7.77	26.27	14.94	4.12	NA	4.12	12.87	Very High	14.23
	Toll Rates	-5									
	Increase in O&M Cost	0									
Scenario-4	Volume of Traffic	10	16.43	38.48	24.25	5.74	NA	5.74	14.87	Very High	18.16
	Toll Rates	5									